What is this project?

The big idea in simple words

Imagine you run a shop. You have customer info in 5 different notebooks — one for purchases, one for complaints, one for website visits, one for emails, one for loyalty points. This project brings ALL of that into ONE place, then uses smart math to group customers and predict what they'll do next.

> THINK OF IT LIKE THIS
>
> You're building a 'super file' for every customer — one card that shows everything about them — then using that to understand who they are and what they need.

What you actually build / do:

- ◯ Combine data from different places into one system
- ◯ Clean and fix messy or duplicate data
- ◯ Group customers by behavior (e.g. loyal buyers vs one-time buyers)
- ◯ Predict which customers might leave soon
- ◯ Show all of this in a nice dashboard

Tools to use:

Python | SQL | Any database | Spreadsheet data

> Resume / interview tip
>
> On your resume, say: 'Built a system that unifies customer data from 5 sources and segments customers using ML.' This immediately sounds impressive because it's a real business problem.

Step 1 — Collect the data

Gather all customer info

Just like a detective collects clues, you collect data about customers from different places. Each place gives you a different piece of the puzzle.

> SIMPLE EXAMPLE
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> Website visits = what pages did they look at? Orders = what did they buy and when? Support tickets = did they ever complain? Emails = did they open your newsletter? Each one tells you something different about the customer.

What you actually build / do:

- ◯ Download or export data from a fake CRM (use free tools like HubSpot free tier or mock data)
- ◯ Create fake order/transaction records using Python's Faker library
- ◯ Simulate website clicks as a CSV file (page visited, time, user ID)
- ◯ Make a support ticket table (customer ID, issue, date, rating)
- ◯ Join all of them using a common field — the customer's ID or email

Tools to use:

Python Faker | CSV files | Pandas | SQLite or PostgreSQL

> Resume / interview tip
>
> Tell recruiters: 'Ingested and unified data from 5 sources including CRM, transactions, and behavioral events.' You don't need real company data — synthetic (fake) data is totally fine for portfolios.

Step 2 — Clean & store the data

Fix messy data and save it properly

Real data is always messy. Some customers appear twice. Some names are misspelled. Some fields are empty. Your job is to clean it up and store it in a nice, organized way.

> THINK OF IT LIKE THIS
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> Imagine getting 5 messy handwritten lists and typing them all into one clean spreadsheet — removing duplicates, fixing spelling, filling in blanks. That's data cleaning.

What you actually build / do:

- ◯ Remove duplicate customer records (same person, different spelling)
- ◯ Fill in or flag missing values (e.g. missing email address)
- ◯ Standardize formats (all dates as YYYY-MM-DD, all phones as +91-XXXXXXXXXX)
- ◯ Create one master 'customer table' with one row per customer
- ◯ Save it in a database (SQLite for simple, PostgreSQL for impressive)

Tools to use:

Pandas | dbt (tool that cleans data using SQL) | PostgreSQL | Great Expectations (checks data quality)

> Resume / interview tip
>
> 'Used dbt to build a clean, tested data pipeline' — dbt is a tool every data job asks for. Even basic usage is a huge green flag for recruiters.

Step 3 — Build customer profiles

Create one complete profile per customer

Now that your data is clean, build a single 'profile card' for every customer. This card shows everything: how often they buy, how much they spend, when they last bought, if they've complained, etc.

> THINK OF IT LIKE THIS
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> It's like a school report card, but for a customer. One page that shows all the important numbers. When a sales person looks at it, they instantly know who this customer is.

What you actually build / do:

- ◯ Calculate 'days since last purchase' for each customer
- ◯ Count total number of purchases per customer
- ◯ Sum total money spent per customer
- ◯ Count how many support tickets they raised
- ◯ Calculate their average product rating (if they give reviews)
- ◯ Store all of this as one row per customer in a 'customer profile' table

Tools to use:

SQL (GROUP BY, SUM, COUNT) | Pandas | dbt models

> Resume / interview tip
>
> 'Built a unified customer profile combining behavioral, transactional, and support data.' This is the core of Customer 360 — saying this in an interview shows you understand the concept deeply.

Step 4 — Group customers (Segmentation)

Sort customers into groups automatically

This is the ML (machine learning) part. You let the computer automatically find groups of similar customers — without you telling it what the groups should be. It figures it out on its own by looking at patterns.

> SIMPLE EXAMPLE
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> You give the computer data about 10,000 customers. It comes back and says: 'Group A = people who buy a lot and recently. Group B = people who bought once long ago. Group C = big spenders who complain a lot.' You didn't define these groups — the algorithm found them.

What you actually build / do:

- ◯ First, do RFM scoring: score every customer on Recency, Frequency, and Money spent (1–5 score each)
- ◯ Then use K-Means clustering: tell Python 'find 5 natural groups in this data'
- ◯ Look at each group and give it a name that makes business sense
- ◯ Visualize the groups on a scatter plot so it looks good in your portfolio
- ◯ Save each customer's group label back to their profile

Tools to use:

Scikit-learn (K-Means) | Matplotlib / Seaborn (charts) | Pandas

> Resume / interview tip
>
> 'Applied RFM scoring and K-Means clustering to segment 100k customers into 6 behavioral groups.' RFM is a well-known industry technique — naming it shows business knowledge, not just coding.

Step 5 — Predict churn

Predict which customers will leave

Churn means a customer stops buying from you. This step builds a model that looks at a customer's behavior and says 'this person has a 78% chance of leaving in the next 30 days.' The business can then reach out to save them.

> THINK OF IT LIKE THIS
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> It's like a doctor looking at your test results and saying 'if you don't change your diet, there's a 70% chance you'll have a problem in 6 months.' The model gives an early warning so someone can act.

What you actually build / do:

- ◯ Label past customers as 'churned' (stopped buying after 90 days) or 'active'
- ◯ Use their profile features (recency, frequency, complaints) as inputs
- ◯ Train an XGBoost model to predict churn — it's the best simple ML model for this
- ◯ Check how accurate it is (aim for 80%+ accuracy)
- ◯ Show a chart of which features matter most (feature importance chart)
- ◯ Track the experiment using MLflow so you can show version history

Tools to use:

XGBoost | Scikit-learn | MLflow (tracks experiments) | Matplotlib

Resume / interview tip

'Built a churn prediction model using XGBoost achieving 83% accuracy, tracked with MLflow.' MLflow shows you care about production ML, not just notebook experiments — huge resume differentiator.

Step 6 — Build the dashboard

Show everything in a nice visual app

You've done all the hard work. Now make it easy for anyone to see and use. Build a simple web app where you can look up any customer, see their segment, see their churn risk, and browse all the groups.

> THINK OF IT LIKE THIS
>
> Think of it like a Wikipedia page for a customer. You type their name, and you see: their full profile, what group they belong to, their churn risk score, and recommendations for how to treat them.

What you actually build / do:

- [ ] Use Streamlit (a Python tool) to build the app — no web design skills needed
- [ ] Add a search box: type a customer ID and see their full profile
- [ ] Add a segments overview page: show all 6 groups with sizes and key stats
- [ ] Add a churn risk page: show the top 50 customers most likely to leave
- [ ] Add one chart per segment showing average spend, frequency, recency
- [ ] Take screenshots — these are your portfolio images

Tools to use:

Streamlit | Plotly (interactive charts) | Pandas

> Resume / interview tip
>
> 'Built an interactive Streamlit dashboard for exploring customer segments and churn risk.' A working demo URL or screenshots in your portfolio is 10x more powerful than code alone.

Step 7 — Polish for your portfolio

Make it look great to employers

The project is done. Now present it in a way that gets you shortlisted. Recruiters spend 30 seconds on a project — make those 30 seconds count.

> THINK OF IT LIKE THIS
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> Even a great restaurant needs a good menu. Your project is the food — but you also need to present it well. A clean README, a short video, and a clear one-line description are your menu.

What you actually build / do:

- [ ] Write a clear README.md: what the project does, why it matters, how to run it
- [ ] Add a simple architecture diagram (the one above!) to your README
- [ ] Record a 2-minute screen recording walking through the dashboard
- [ ] Push everything to GitHub with clean commit messages
- [ ] Write a LinkedIn post: 'I built X that does Y, here's what I learned' with screenshots
- [ ] In your resume: 'Customer 360 Analytics — unified 5 data sources, built ML segmentation (6 groups), predicted churn with 83% accuracy, built Streamlit dashboard'

Tools to use:

GitHub | Loom (screen recording) | LinkedIn | Resume

Resume / interview tip

The one-liner that gets shortlisted: 'End-to-end customer analytics system — data pipeline, ML segmentation, churn prediction, and interactive dashboard.' Every word in that sentence maps to a skill recruiters search for.